

Exploring the ESP32 as an Attack Platform

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Abstract—This project explores the capabilities of the ESP32 microcontroller as a portable, low-cost offensive security platform. While ARP spoofing is a well-documented vulnerability, its implementation on resource-constrained embedded devices raises important questions about practical attack capabilities. Rather than simply demonstrating individual attacks, this research progressively evaluates the device’s capabilities across multiple vectors including ARP spoofing, DNS spoofing, deauthentication attacks, and rogue access point deployment with captive portals. Using an ESP32 development board with Wireshark for traffic analysis in a controlled network environment, we measure success rates and stealth factors by varying packet rates to avoid detection. Expected outcomes include a working proof-of-concept ESP32 firmware that chains multiple attacks, quantitative data on detection thresholds, and recommendations for defending against ESP32-based network attacks.

I. INTRODUCTION

The ESP32 microcontroller, present in over a billion IoT devices worldwide, represents an intriguing attack surface. While ARP spoofing is a well-documented vulnerability, its implementation on resource-constrained embedded devices raises important questions about practical attack capabilities. This project aims to explore the limits of what an ESP32 can achieve as a portable, low-cost offensive security platform.

The central question is: “How far can we go with an ESP32?” Rather than simply demonstrating individual attacks, this project aims to progressively evaluate the device’s capabilities across multiple vectors. Starting with ARP spoofing, we aim to incrementally add attack layers: DNS and DHCP spoofing, deauthentication attacks to force disconnections, and rogue access point deployment with captive portals for credential harvesting.

The project will use an ESP32 development board, with Wireshark for traffic analysis and a controlled network of victim machines. We will measure success rates, as well as stealth factors (varying packet rates to avoid detection).

We expect to produce a working proof-of-concept ESP32 firmware that chains multiple attacks, quantitative data on detection thresholds, and recommendations for defending against ESP32-based network attacks.

II. RELATED WORK

Several open-source projects have demonstrated individual attack vectors using the ESP32. The ESP32-ARP-Spoof repository by Tanjib provides a MicroPython implementation of ARP spoofing that can intercept unencrypted traffic within an SSID and cause connectivity issues for victim devices [1]. The author notes that sending 500 packets rapidly might trigger

intrusion detection systems, suggesting a stealthier approach of sending ARP packets at intervals to mimic normal behavior.

Other projects have explored different attack vectors. The MIM-attack-ESP32 repository implements a captive portal for credential harvesting [2], while IoT_Flipper provides a comprehensive toolkit combining multiple offensive capabilities [3]. Academic research has also examined ESP32 vulnerability during OTA updates, demonstrating successful ARP spoofing and TCP SYN flooding attacks [4].

REFERENCES

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